

Tutorial

Indian Institute of Technology Jodhpur
Mathematics-I (MAL1010)

Instructors: Dr. Subhash Bhagat, Dr. Md Abu Talhamainuddin Ansary

1. Evaluate the double integral

$$\iint_R (x + y) dA$$

where R is the triangular region with vertices at $(0, 0)$, $(1, 0)$, $(1, 1)$.

2. Evaluate the double integral by changing the order of integration:

$$\int_0^1 \int_y^1 \frac{x}{x^2 + 1} dx dy$$

3. Evaluate the integral

$$\iint_R (x^2 + y^2) dA,$$

where R is the region bounded by the circle $x^2 + y^2 = 4$.

4. Evaluate the volume of the solid bounded by the cylinder $x^2 + y^2 = 4$ and the planes $z = 0$ and $z = x + y$.

5. Evaluate the following integrals

(i)

$$\iint_R (4 - 2y) dA, \quad \text{where } R = [0, 1] \times [0, 1].$$

(ii)

$$\iint_R \sin x \cos y dA, \quad \text{where } R = [0, \frac{\pi}{2}] \times [0, \frac{\pi}{2}].$$

(iii)

$$\int_0^1 \int_0^1 \sqrt{s + t} ds dt.$$

6. Find the volume of the solid S that is bounded by the elliptic paraboloid

$$x^2 + 2y^2 + z = 16,$$

the planes $x = 2$, $y = 2$, and the three coordinate planes.

7. Evaluate

$$\iint_D (x + 2y) dA,$$

where D is the region bounded by the parabolas

$$y = 2x^2 \quad \text{and} \quad y = 1 + x^2.$$

8. Find the volume of the solid bounded by the plane $z = 0$ and the paraboloid

$$z = 1 - x^2 - y^2.$$